

Which Edges Can a GNN Trust?

Baseline results on WikiCS

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1 Summary

2 Setup

Data. WikiCS [?]: 11,701 Wikipedia articles on computer science, 215,603 undirected hyperlink edges, and 10 topic labels. Each node has its raw article text (title plus tokens) and a 300-dimensional averaged GloVe feature vector. Clean edge homophily (the share of edges joining two nodes with the same label) is 0.654. I use the first 3 of the 20 standard train/validation splits (580 training nodes each) and the standard test set of 5,847 nodes. The data was loaded from the original JSON release.

Attack. Fake edges are injected between nodes whose texts look related but whose labels differ. For every node I take its 60 nearest neighbours by SBERT cosine similarity (all-MiniLM-L6-v2 on title plus the first 256 tokens) and keep the non-adjacent pairs with different labels. This gives 201,981 candidate pairs. An attack at budget b samples $b \cdot 215,603$ of them uniformly, for $b \in \{5, 10, 20, 40\}\%$. As a contrast, a *random* attack at $b = 20\%$ samples non-edges uniformly, ignoring text and labels. Because I inject the fake edges myself, every edge has a known ground-truth label (real or fake).

Models. A 2-layer GCN [?] and a 2-layer MLP that ignores edges, both with 64 hidden units, dropout 0.5, Adam (learning rate 0.01, weight decay 5×10^{-4}), up to 300 epochs, and early stopping on validation loss (patience 50). Node features are the GloVe vectors. The GCN at budget 0% is also the *oracle defense*: it is what any method that removed exactly the fake edges would achieve.

Node metrics. Test accuracy, negative log-likelihood (NLL), expected calibration error (ECE, 15 bins) and Brier score. I also split test accuracy by whether a node is an endpoint of at least one fake edge (*touched*) or not.

Edge-detection baselines. Each method gives every edge of the attacked graph a suspicion score, higher meaning more likely fake. Scores are compared with the ground truth by AUROC and AUPRC, with fake edges as the positive class.

- **Text only:** $1 - \cos$ of the two endpoints' SBERT embeddings, and the same for their GloVe features. These stand in for the text prior until the LLM prior is added.

- **Structure only:** 1− Jaccard overlap of the endpoints’ neighbourhoods, and the Jensen–Shannon divergence between the GCN’s predicted class distributions at the two endpoints (the GCN trained on the attacked graph).
- **Naive mix:** the unweighted average of the rank-normalised text and structure scores. This is the “hand-made rule” that the Bayesian model should beat.

3 Results

3.1 Node classification

Model	Attack	Acc. $\uparrow$	NLL $\downarrow$	ECE $\downarrow$	Brier $\downarrow$	Acc. touched	Acc. untouched
MLP (no edges)	–	0.738 ± 0.010	0.781 ± 0.018	0.021 ± 0.011	0.371 ± 0.008	–	–
GCN	none	0.794 ± 0.005	0.724 ± 0.015	0.043 ± 0.006	0.310 ± 0.009	–	0.794 ± 0.005

Table 1: Test-set node classification on WikiCS, mean $\pm$ std over 3 splits. “Touched” nodes are endpoints of at least one fake edge.

3.2 Fake-edge detection

	Text only		Structure only		Naive mix	
Attack	SBERT	GloVe	Jaccard	GCN JS	SBERT+GCN	SBERT+Jacc.

Table 2: Fake-edge detection AUROC (fake = positive; 0.5 = chance, below 0.5 = ranks fakes as *more* trustworthy). Mean over 3 splits; best per row in bold.

	Text only		Structure only		Naive mix	
Attack	SBERT	GloVe	Jaccard	GCN JS	SBERT+GCN	SBERT+Jacc.

Table 3: Fake-edge detection AUPRC. The chance level equals the share of fake edges: 0.048, 0.091, 0.167 and 0.286 at budgets 5, 10, 20 and 40%.

4 What this means for the hypotheses

5 Limitations and next steps

References

- [1] P. Mernyei and C. Cangea. Wiki-CS: A Wikipedia-based benchmark for graph neural networks. *ICML Workshop on Graph Representation Learning*, 2020.
- [2] T. N. Kipf and M. Welling. Semi-supervised classification with graph convolutional networks. *ICLR*, 2017.